13. Symbol of lead is ___________.
(a) Pd (b) W (c) Pb (d) Sn

14. Which among the following is a salt of 'ic' acid?
(a) $NaNO_2$ (b) Na_2SO_3
(c) Na_2SO_4 (d) $NaClO_2$

15. For which among the following reactions sound energy is required?
(a) $2KClO_3 \rightarrow 2KCl + 3O_2$ (b) $C_2H_2 \rightarrow 2C + H_2$
(c) $2AgCl \rightarrow 2Ag + Cl_2$ (d) $2H_2O \rightarrow 2H_2 + O_2$

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Heating of ammonium chloride involves only physical change.

   Reason (R): Ammonium chloride on heating decomposes to ammonia and hydrogen chloride.

   (a) Both A and R are true and R is the correct explanation of A.

   (b) Both A and R are true, but R is not the correct explanation of A.

   (c) A is true and R is false.

   (d) A is false and R is true.

2. **Assertion (A):** Criss-cross method is used to derive the formula of the compounds.

   Reason (R): The molecule of any compound is neutral.

   (a) Both A and R are true and R is the correct explanation of A.

   (b) Both A and R are true, but R is not the correct explanation of A.

   (c) A is true and R is false.

   (d) A is false and R is true.

3. Arrange the following negative radicals in the descending order of the magnitude of charges.

   (A) Fluoride (B) Oxide

   (C) Nitride (D) Carbide

   (a) DCBA (b) DCAB

   (c) DCAB (d) CDAB

4. A, B, C and D are divalent metals which show the following reactions.

   $ASO_4 + B \rightarrow$ No reaction
   (aq)

   $ASO_4 + C \rightarrow CSO_4 + A.$
   (aq)

   $BSO_4 + D \rightarrow$ No reaction
   (aq)

   Arrange A, B, C and D in the descending order of their reactivity.

   (a) CABD (b) CADB

   (c) ACBD (d) ACDB

Space for rough work

5. Match the entries of Column A with those of Column B.

Column A (Formula)	Column B (Total Positive Charge)
(i) Zinc nitride	(A) 2
(ii) Plumbic chloride	(B) 3
(iii) Sodium phosphate	(C) 5
(iv) Calcium nitrate	(D) 4
	(E) 1
	(F) 6

(a) (i) → (D); (ii) → (F); (iii) → (B); (iv) → (A)
(b) (i) → (D); (ii) → (F); (iii) → (A); (iv) → (B)
(c) (i) → (F); (ii) → (D); (iii) → (A); (iv) → (B)
(d) (i) → (F); (ii) → (D); (iii) → (B); (iv) → (A)

6. Identify *a* and *b* from the following equation: $a\ HNO_3 + S \rightarrow H_2SO_4 + 6NO_2 + b\ H_2O$.
(a) $a = 6, b = 2$ (b) $a = 6, b = 3$
(c) $a = 6, b = 4$ (d) $a = 4, b = 2$

7. Molecular mass of phosphate of divalent metal is 262. Find the atomic mass of the metal.
(a) 24 (b) 27 (c) 40 (d) 23

8. Which among the following is an analysis reaction?
(a) $2Na + Cl_2 \rightarrow 2NaCl$ (b) $2NaCl \rightarrow 2Na + Cl_2$
(c) $CaCO_3 \rightarrow CaO + CO_2$ (d) $2HOCl \rightarrow 2HCl + O_2$

9. 40 g of sulphur on treatment with sufficient amount of oxygen gives how many grams of SO_3?
$[S + O_2 \rightarrow SO_3]$
(a) 100 g (b) 140 g (c) 75 g (d) 85 g

10. Which among the following metals shows variable valency?
(a) Fe (b) Mg (c) Zn (d) Al

11. The ratio of weight of nitrogen combining with fixed weight of oxygen in N_2O, NO, and N_2O_5 is __________.
(a) 10:5:4 (b) 10:5:2 (c) 10:5:1 (d) 10:5:3

12. The atomic mass of an element is 24. What is the element?
(a) Mg (b) S (c) Na (d) Cr

13. Greek name of antimony is__________.
(a) *kalium* (b) *natrium*
(c) *stannum* (d) *stibnum*

14. Identify the binary acid among the following.
(a) Nitric acid (b) Hydrochloric acid
(c) Phosphoric acid (d) Sulphuric acid

15. Identify the photochemical reaction among the following.
(a) Photosynthesis (b) Respiration
(c) Rusting (d) Combustion

Space for rough work

Answer Keys

Assessment Test I

1. (d) **2.** (a) **3.** (a) **4.** (b) **5.** (c) **6.** (a) **7.** (a) **8.** (a) **9.** (b) **10.** (b)
11. (d) **12.** (b) **13.** (a) **14.** (c) **15.** (c)

Assessment Test II

1. (a) **2.** (b) **3.** (d) **4.** (c) **5.** (a) **6.** (a) **7.** (c) **8.** (a) **9.** (b) **10.** (a)
11. (d) **12.** (c) **13.** (d) **14.** (c) **15.** (c)

Assessment Test III

1. (a) **2.** (d) **3.** (b) **4.** (b) **5.** (b) **6.** (d) **7.** (b) **8.** (c) **9.** (b) **10.** (d)
11. (b) **12.** (c) **13.** (c) **14.** (c) **15.** (b)

Assessment Test IV

1. (d) **2.** (a) **3.** (a) **4.** (a) **5.** (d) **6.** (a) **7.** (a) **8.** (b) **9.** (a) **10.** (a)
11. (b) **12.** (a) **13.** (d) **14.** (b) **15.** (a)

Air and Oxygen 4

Reference: Coursebook - IIT Foundation Chemistry Class 8; Chapter - Air and Oxygen; Page number - 4.1–4.21

Assessment Test I

Time: 30 min.

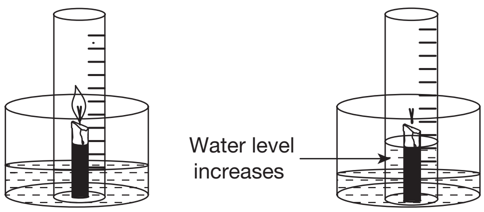
Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. In which of the following layers of atmosphere, the temperature increases with the increase in altitude?
   (a) Troposphere (b) Stratosphere
   (c) Mesosphere (d) Thermosphere

2. The height of the mercury column in barometer is 19 cm at a certain place. Calculate the pressure in Pascal in that place. (1 atm = 101.3 kilopascals)
   (a) 39,925 (b) 1,01,292 (c) 25,323 (d) 30,353

3. Which of the following is the conclusion drawn from the experiment given below?



   (a) Nitrogen is the major component of air.
   (b) CO_2 is present in air.
   (c) O_2 is one of the components of air.
   (d) One-fifth of the volume of air is oxygen.

4. Which of the following processes does not consume O_2?
   (a) Photosynthesis (b) Respiration
   (c) Burning of fuel (d) Rusting

5. Which of the following is an example of rapid combustion?
   (a) Burning of white phosphorus
   (b) Burning of petrol
   (c) Burning of crackers
   (d) Burning of sodium in moist air

Space for rough work

6. The innermost zone of a candle flame is ________.
   (a) the coldest zone (b) the hottest zone
   (c) the luminous zone (d) the non-luminous zone

7. Which of the following metals can form a mixed oxide on reaction with O_2?
   (a) Al (b) Cu (c) Ag (d) Fe

8. Identify the products formed when a mixture of NH_3 and O_2 in 4:5 ratio is passed over platinum gauze at 800°C.
   (a) NO and H_2O (b) N_2 and H_2O
   (c) NO_2 and H_2 (d) NO and H_2

9. Identify the correct product.
   (a) $H_2S + O_2 \rightarrow H_2O + SO_2$ (b) $H_2S + O_2 \rightarrow S + H_2O$
   (c) $H_2S + O_2 \rightarrow H_2SO_4$ (d) $H_2S + O_2 \rightarrow SO_2 + H_2$

10. For making surgical instruments, which of the following processes can be adopted for prevention of corrosion of iron?
    (a) Galvanization (b) Tinning
    (c) Electroplating (d) Alloying

11. **Assertion (A):** Oxygen is used for the combustion of fuel.
    Reason (R): Combustion is an exothermic reaction.
    (a) Both A and R are true and R is the correct explanation of A.
    (b) Both A and R are true, but R is not the correct explanation of A.
    (c) A is true and R is false.
    (d) A is false and R is true.

12. **Assertion (A):** Air supports combustion.
    Reason (R): Air is a mixture of different gases.
    (a) Both A and R are true and R is the correct explanation of A.
    (b) Both A and R are true, but R is not the correct explanation of A.
    (c) A is true and R is false.
    (d) A is false and R is true.

13. Arrange the following measures/equipment for controlling pollution in the order of the pollutants mentioned below.
    (A) Usage of scrubber in industries
    (B) Controlling deforestation
    (C) Developing a suitable substituent for the refrigerant used in A/C, refrigerator, etc.
    (D) Usage of electrostatic precipitators in industries

(Suspended particulate matter, CFCs, oxides of sulphur and excess CO_2 in atmosphere)

(a) DACB (b) BCDA (c) DCBA (d) DCAB

14. Arrange the following products of oxidation of non-metals in the order of their description given below.

(A) P_2O_5 (B) NO_2 (C) CO_2 (D) NO

Description:

(i) The gas produces a brown coloured gas on oxidation.

(ii) The gas used in fire extinguishers.

(iii) A brown coloured gas

(iv) A dense white fume

(a) DACB (b) DCBA (c) BDCA (d) CDBA

15. Match the entries of Column A with those of Column B.

Column A (Formation of O_2 From Different Metallic Oxides by Heating)	Column B (Observation)
(i) PbO_2	(A) Reddish brown residue turns yellow on cooling.
(ii) HgO	(B) Silvery white globules are formed.
(iii) Pb_3O_4	(C) Mirror like surface near the cooler parts of the test tube.
(iv) Ag_2O	(D) Chocolate brown coloured substance turns yellow.

(a) (i) → (D); (ii) → (B); (iii) → (C); (iv) → (A)

(b) (i) → (D); (ii) → (C); (iii) → (B); (iv) → (A)

(c) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (C)

(d) (i) → (D); (ii) → (C); (iii) → (A); (iv) → (B)

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. Which of the following layers of the atmosphere exists in plasma state (ionic state)?
   (a) Troposphere (b) Stratosphere
   (c) Mesosphere (d) Thermosphere
2. At certain altitude, the atmospheric pressure is 0.05 atm. What would be the length of the mercury column in the barometer?
   (a) 3.8 cm (b) 7.6 cm
   (c) 1.9 cm (d) 2.5 cm
3. The percentage of which of the following components of air always varies from one place to the other?
   (a) O_2 (b) N_2
   (c) Water vapour (d) CO_2
4. Which of the following processes does not produce O_2?
   (a) Heating of red lead (b) Photosynthesis
   (c) Mixing of MnO_2 and H_2O_2 (d) Combustion
5. Respiration is a _________.
   (a) spontaneous combustion (b) rapid combustion
   (c) slow combustion (d) explosion
6. Which of the following is not the characteristic feature of the outermost zone of the flame of a candle?
   (a) This zone is non-luminous. (b) Carbon particles burn here.
   (c) It does not produce any soot. (d) It is the hottest zone.
7. Which of the following metals can protect itself from further oxidation?
   (a) Ca (b) Al (c) Cu (d) K
8. Identify the products formed when ammonia is burnt in limited supply of O_2.
   (a) N_2O and H_2O (b) NO and H_2O
   (c) NO_2 and H_2O (d) N_2 and H_2O
9. Identify the correct product.
   (a) $FeS_2 + O_2 \rightarrow FeO + SO_2$ (b) $FeS_2 + O_2 \rightarrow FeO + S$
   (c) $FeS_2 + O_2 \rightarrow Fe_2O_3 + S$ (d) $FeS_2 + O_2 \rightarrow Fe_2O_3 + SO_2$
10. For manufacturing the containers for storing edible materials like oils, fruit juices, etc., which of the following methods can be adopted for prevention of corrosion of iron?

Space for rough work

(a) Electroplating (b) Galvanizing
(c) Tinning (d) Alloying

11. **Assertion (A):** Oxyacetylene flame is used for welding purposes.
Reason (R): Combustion of acetylene is an example of rapid combustion.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

12. **Assertion (A):** When air is passed through lime water, it turns milky.
Reason (R): Constituents of a mixture retain their properties.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

13. Arrange the following pollutants in the order of their effects mentioned below.
(A) Fly ash (B) CFCs
(C) SO_2 (D) Ozone
(Acid rain, global warming, bronchitis and other respiratory disorders, and ozone depletion)
(a) DCAB (b) ACBD
(c) CDAB (d) BCDA

14. Arrange the following oxides of metals in the order of their description given below.
(A) Al_2O_3 (B) CuO
(C) CaO (D) Fe_3O_4
Description:
(i) This oxide is black in colour.
(ii) This oxide protects the metal from further oxidation.
(iii) This brown coloured powdery oxide is formed with a cracking noise when the metal is heated at high temperature.
(iv) This white coloured oxide is formed on slight heating of the metal.
(a) BADC (b) DCBA
(c) ACBD (d) CDAB

Space for rough work

15. Match the entries of Column A with those of Column B.

Column A (Compounds Used for the Formation of O_2)	Column B (Related Features of the Reaction Involved)
(i) H_2O_2	(A) MnO_2 reduces the melting point as well as decomposition temperature of the reactant.
(ii) $KMnO_4$	(B) Metallic nitrite is the solid product.
(iii) $KClO_3$	(C) No heating is required.
(iv) $NaNO_3$	(D) MnO_2 is one of the solid products.

(a) (i) → (C); (ii) → (A); (iii) → (D); (iv) → (B)
(b) (i) → (C); (ii) → (B); (iii) → (A); (iv) → (D)
(c) (i) → (C); (ii) → (D); (iii) → (A); (iv) → (B)
(d) (i) → (C); (ii) → (D); (iii) → (B); (iv) → (A)